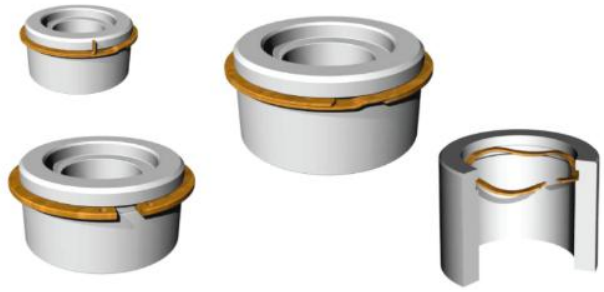
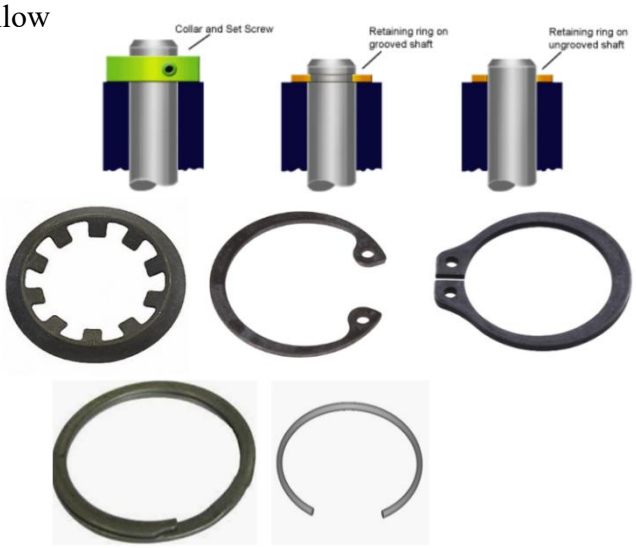
Lesson 05: Shafts, Key, O-ring, Retainer rings and Coupling

AXIAL LAYOUT OF COMPONENTS:



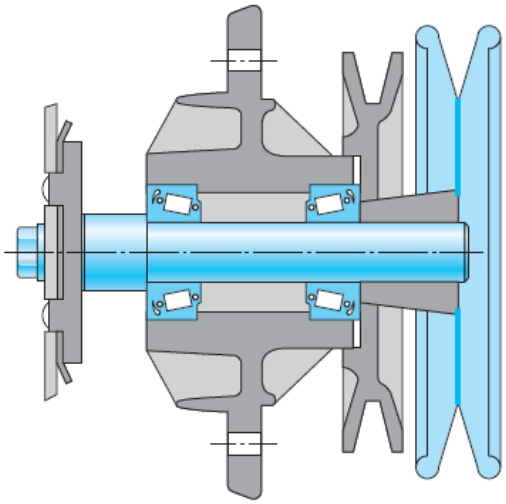
- Only two bearings should be used in most cases.
- For extremely long shafts carrying several load-bearing components, it may be necessary to provide more than two bearing supports.
- In this case, particular care must be given to the alignment of the bearings,
- Shafts should be kept short to minimize bending moments and deflections.
- Some axial space between components is desirable to allow for lubricant flow and to provide access space for disassembly of components with a puller.
- Load bearing components should be placed near the bearings, again to minimize the bending moment at the locations that will likely have stress concentrations, and to minimize the deflection at the load-carrying components.
- The components must be accurately located on the shaft to line up with other mating components,
- A shoulder provides a solid support to minimize deflection and vibration of the component. As well as to locate bearing in place.
- Sometimes when the magnitudes of the forces are reasonably low, shoulders can be constructed with retaining rings in grooves, sleeves between components, or clamp-on collars.
- In cases where axial loads are very small, it may be feasible to do without the shoulders entirely, and rely on press fits, pins, or collars with setscrews to maintain an axial location

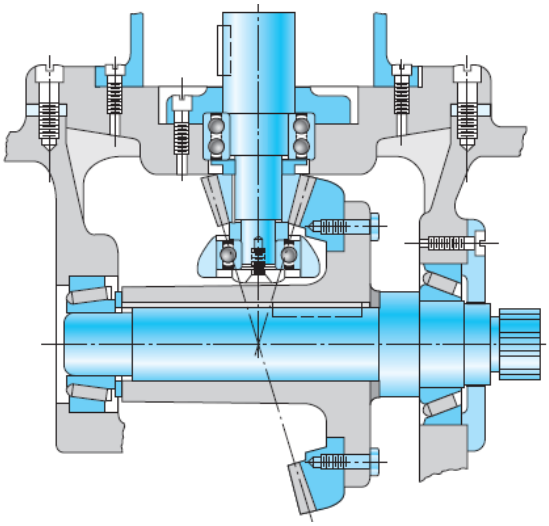




SUPPORTING AXIAL LOADS:

- In cases where axial loads are not trivial, it is necessary to provide a means to transfer the axial loads into the shaft, then through a bearing to the ground.
- This will be particularly necessary with helical or bevel gears, or tapered roller bearings, as each of these produces axial force components.
- Often, the same means of providing axial location, e.g., shoulders, retaining rings, and pins, will be used to also transmit the axial load into the shaft.





Providing for Torque Transmission:

- Most shafts serve to transmit torque from an input gear, pulley or coupling, through the shaft, to an output gear or pulley.
- The shaft must be sized to support the torsional stress and torsional deflection.
- It is also necessary to provide a means of transmitting the torque between the shaft and the gears. Common torque-transfer elements are:

    - Keys
    - Splines
    - Setscrews
    - Coupling
    - Pin
    - Press or shrink fits
    - Tapered fits

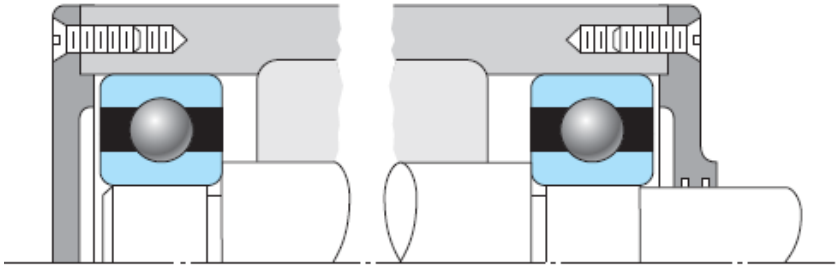
- In addition to transmitting the torque, many of these devices are designed to fail if the torque exceeds acceptable operating limits,

ASSEMBLY AND DISASSEMBLY:

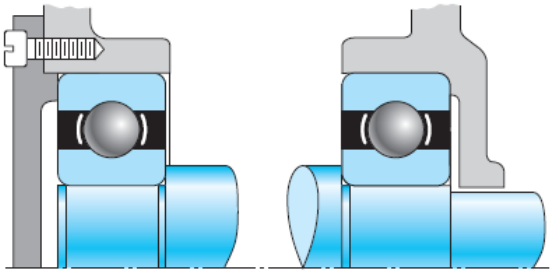
- When design a shaft, Consideration should be given to:
  - the method of assembling the components onto the shaft,
  - and the shaft assembly into the frame.

- This generally requires the largest diameter in the center of the shaft, with progressively smaller diameters towards the ends to allow components to be slid on from the ends.
- If a shoulder is needed on both sides of a component, one of them must be created by such means as a retaining ring or by a sleeve between two components.

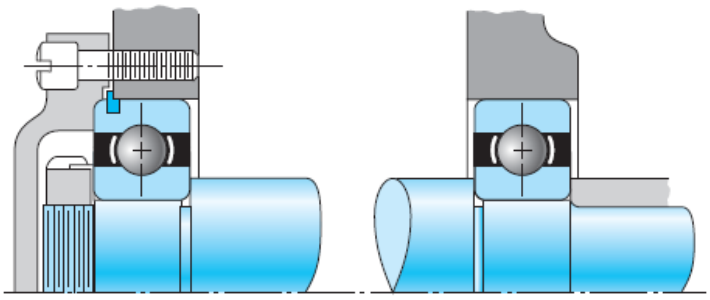
- Arrangement showing bearing inner rings press-fitted to shaft while outer rings float in the housing.



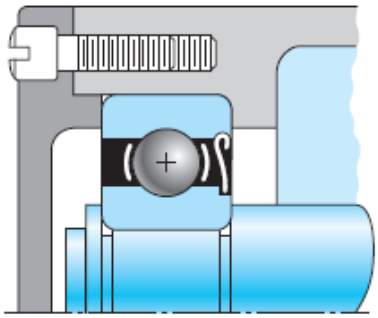
- Similar to the arrangement to the previoues example except that the outer bearing rings are preloaded



- In this arrangement the inner ring of the left-hand bearing is locked to the shaft between a nut and a shaft shoulder. The snap (retainer) ring in the outer race is used to positively locate the shaft assembly in the axial direction..



- In this case the inner ring is secured to the shaft using a snap ring.
  Note the use of a shield to prevent dirt generated from within the machine from entering the bearing



MISCELLANEOUS SHAFT COMPONENTS:

Setscrews:

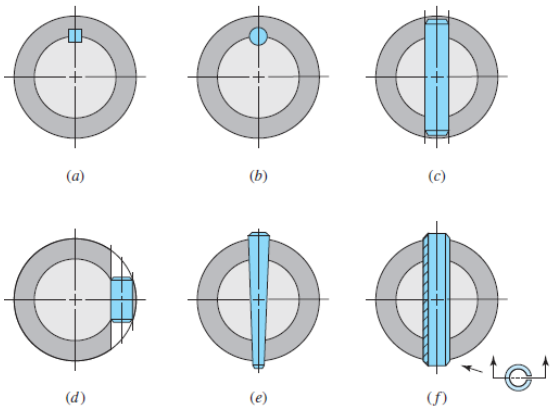
- Unlike bolts and cap screws, which depend on tension to develop a clamping force, the setscrew depends on compression to develop the clamping force.
- Clamping force is due to frictional resistance of the contacting portions of the collar and shaft as well as any slight penetration of the setscrew into the shaft.

Keys and Pins:

- Keys and pins are used on shafts to secure rotating elements, such as gears, pulleys, or other wheels.
- Keys are used to enable the transmission of torque from the shaft to the shaft-supported element.
- Pins are used for axial positioning and for the transfer of torque or thrust or both,

Types of keys:

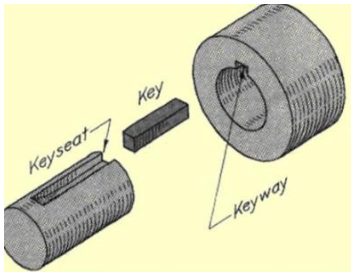
- Square key;
- Round key;
- (c and d) round pins;
- (e) taper pin;
- (f) split tubular spring pin.

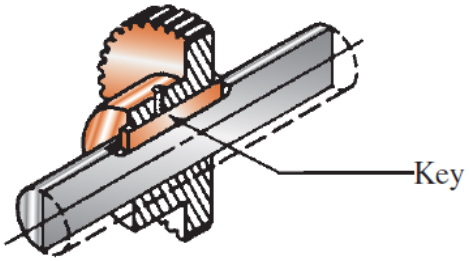


The pins in parts (e) and (f) are shown longer than necessary, to illustrate the chamfer on the ends, but their lengths should be kept smaller than the hub diameters to prevent injuries due to projections on rotating parts.

Keys are Often attached to rotating parts or rotating shaft in order to transmit power.

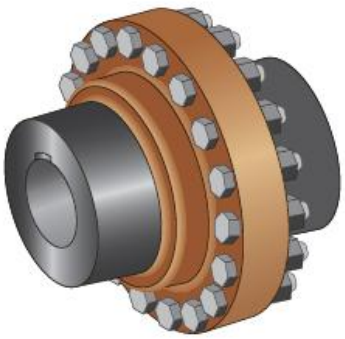
- A key is a machinery component placed at the interface between a shaft and the hub of a power-transmitting element.
- A key to transmit torque between a shaft and associated hub (gear, pully, flange, etc)
- A coupling to transmit torque between TWO shafts.

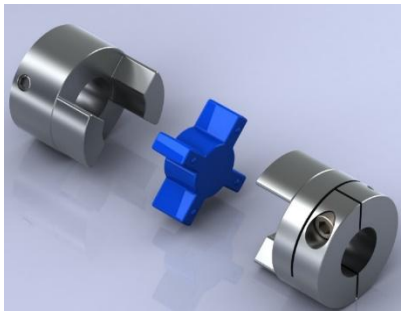






☐ With Slit ☐ Disc Type ☐ Oldham Type ☐ Jaw Type ☐ Bellows Type

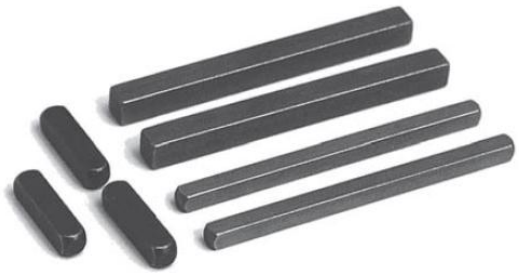




HUMBER

MECH 214 Mechanical Components Design

Types of keys:



Square and Rectangular or Parallel Keys

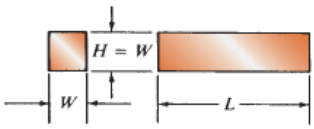
- Keys been standardized to base on shaft size to shaft load capabilities,
- Length of keys are not standardized yet,
- Normally Require set screw
- Both the square and the rectangular keys are referred to as parallel keys because the top and bottom and the sides of the key are parallel.



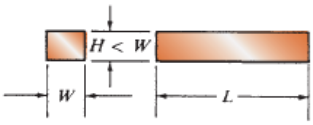
Square and rectangular Key Size vs. Shaft Diameter:

U.S. inch sizes				SI metric sizes			
Nominal shaft diameter		Key dimensions		Nominal shaft diameter		Key dimensions	
Over (in)	to-including (in)	Width, *W* (in)	Height, *H* (in)	Over (mm)	to-including (mm)	Width, *W* (mm)	Height, *H* (mm)
0.3125	0.4375	0.09375	0.09375	6	8	2	2
0.4375	0.5625	0.1250	0.1250	8	10	3	3
0.5625	0.875	0.1875	0.1875	10	12	4	4
0.875	1.250	0.2500	0.2500	12	17	5	5
1.250	1.375	0.3125	0.3125	17	22	6	6
1.375	1.75	0.375	0.375	22	30	8	7
1.75	2.25	0.500	0.500	30	38	10	8
2.25	2.75	0.625	0.625	38	44	12	8
2.75	3.25	0.750	0.750	44	50	14	9
3.25	3.75	0.875	0.875	50	58	16	10
3.75	4.50	1.00	1.00	58	65	18	11
4.50	5.50	1.25	1.25	65	75	20	12
5.50	6.50	1.50	1.50	75	85	22	14
6.50	7.50	1.75	1.50	85	95	25	14
7.50	9.00	2.00	1.50	95	110	28	16
9.00	11.00	2.50	1.75	110	130	32	18
11.00	13.00	3.00	2.00	130	150	36	20
13.00	15.00	3.50	2.50	150	170	40	22
15.00	18.00	4.00	3.00	170	200	45	25
18.00	22.00	5.00	3.50	200	230	50	28
22.00	26.00	6.00	4.00	230	260	56	32
26.00	30.00	7.00	5.00	260	290	63	32
				290	330	70	36
				330	380	80	40
				380	440	90	45
				440	500	100	50

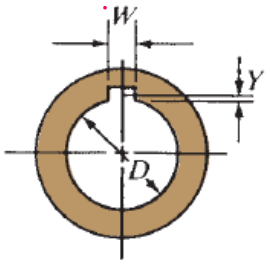
- The key-seats in the shaft and the hub are designed so that exactly one-half of the height of the key.
- The distance Y is the radial distance from the theoretical top of the shaft, before the key-seat is machined, to the top edge of the finished key-seat to produce a key-seat depth of exactly H/2.
- To assist in machining and inspecting the shaft or the hub, the dimensions S and T can be computed and shown on the part drawings.



(b) Square key

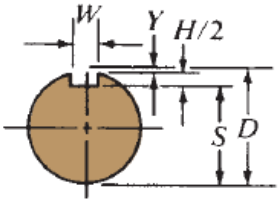


(c) Rectangular key



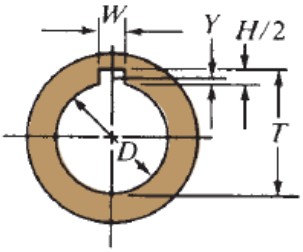
(a) Chordal height

$$Y = \frac{D - \sqrt{D^2 - W^2}}{2}$$



(b) Depth of shaft keyseat

$$S = D - Y - \frac{H}{2} = \frac{D - H + \sqrt{D^2 - W^2}}{2}$$



(c) Depth of hub keyseat

$$T = D - Y + \frac{H}{2} + C = \frac{D + H + \sqrt{D^2 - W^2}}{2} + C$$

Symbols

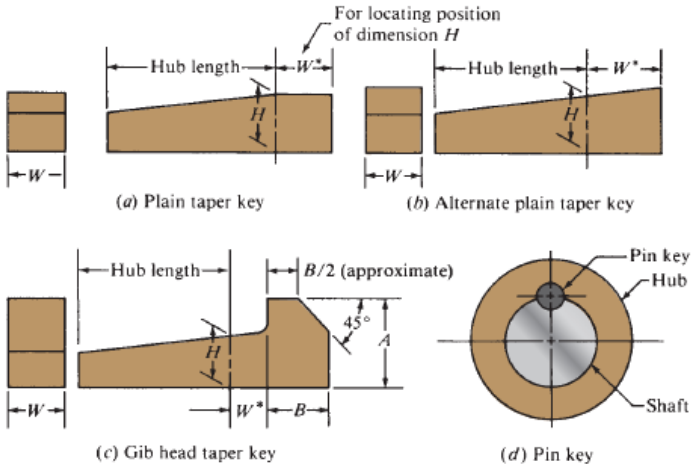
C = Allowance
+ 0.005-in clearance for parallel keys
− 0.020-in interference for taper keys
D = Nominal shaft or bore dia., in
H = Nominal key height, in
W = Nominal key width, in
Y = Chordal height, in

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MECH 214 Mechanical Components Design

Tappered Key:

- Can install after hub (gear) is installed over shaft.
- The taper will generate a compressive contact pressure between the hub and the shaft.
- friction will help transmit torque and provide resistance to axial motion of the hub relative to the shaft.
- Tapered keys do not require set screws.
- Access to both ends of tapered keys are required so that the key can be inserted and driven out when the key is being removed.

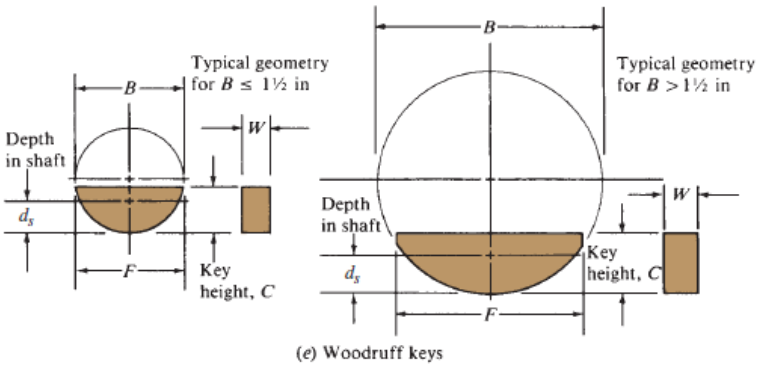


Gib Head Keys:

- A handle on the end to air removing the key with a tool

Woodruff Keys:

- To eliminate the concern about falling key
- Light loading offers ease of assembly



Pin keys

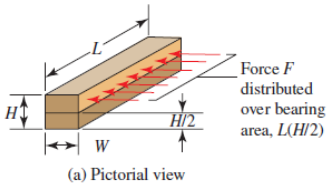
- Low stress concentration (Cutting Keyways create stress concentrations in shafts)

HUMBER

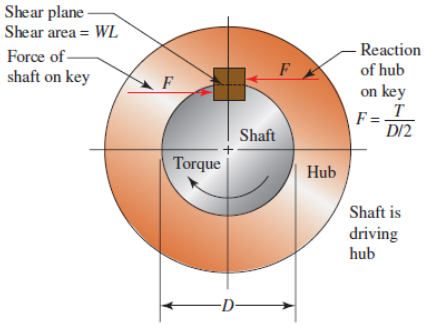
MECH 214 Mechanical Components Design

DESIGN OF PARALLEL KEYS:

Stress analysis to determine required length:



(a) Pictorial view



T = F x (D/2)

or

F = T/(D/2)

Required Length based on Shear Stress:

$$\tau = \frac{F}{A_s} = \frac{T}{(D/2)(WL)} = \frac{2T}{DWL}$$

- In design, we can set the shear stress equal to a design stress in shear for the maximum shear stress theory of failure:

$$\tau_d = 0.5 s_y / N$$

Where N is Design Factor, typically in industrial applications, N = 3 is adequate to accommodate accidental overloads and shock.

Hence, Minimum required length is:

$$L_{\min} = \frac{2T}{\tau_d D W}$$

Material designation	Tensile strength s_u (ksi)	Tensile strength s_u (MPa)	Yield strength s_y (ksi)	Yield strength s_y (MPa)
Carbon steels (SAE)				
1018	64	441	54	372
1035	72	496	39.5	272
1045	91	627	77	531
1095	140	965	83	572
Alloy steels (SAE)				
4140	102	703	90	621
8630	100	690	95	655
Stainless steels (SAE)				
303	90	621	35	241
304	85	586	35	241
316	85	586	35	241
416	75	517	40	276
Aluminum				
6061	18	124	12	83

Required Length based on Bearing Stress:

Failure occurs on the surface with the lowest compressive yield strength.

Design stress for compression is:

$$\sigma_d = s_y/N$$

Where N is Design Factor, typically in industrial applications, N = 3 is adequate to accommodate accidental overloads and shock.

σ_d : Is allowable normal stress.

$$\sigma = \frac{F}{A_c} = \frac{T}{(D/2)(L)(H/2)} = \frac{4T}{DLH}$$

Hence, Minimum required length is:

$$L_{\min} = \frac{4T}{\sigma_d DH}$$

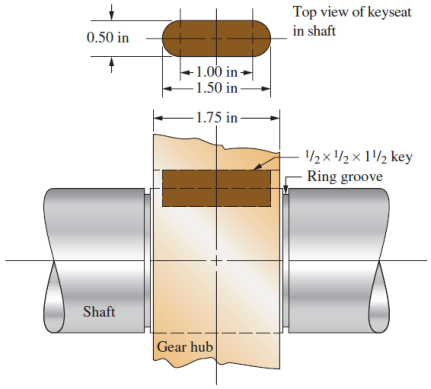
Where: σ_d = normal stress allowable

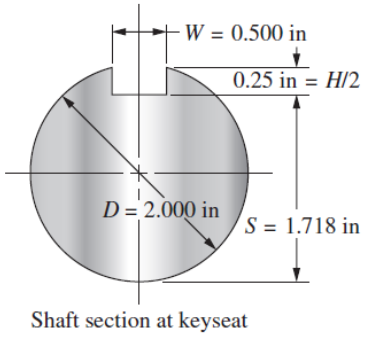
Example:
The portion of a shaft where a gear is to be mounted has a diameter of 2.00 in. The gear transmits 2965 lb. in of torque. The shaft is to be made of SAE 1040 cold-drawn steel. The gear is made from SAE 8650 OQT 1000 steel. The width of the hub of the gear mounted at this location is 1.75 in. Design the key.

Solution:
Material selection is a design decision. (for this example) choose SAE 1018 steel with sy = 54000 psi as listed in the table above.

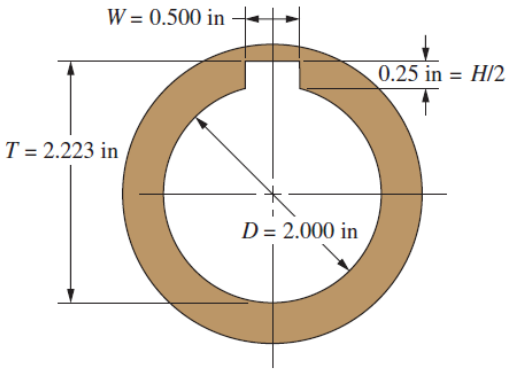
$$L = \frac{4TN}{DWs_y} = \frac{4(2965\ \text{lb}\cdot\text{in})(3)}{(2.00\ \text{in})(0.500\ \text{in})(54\,000\ \text{lb/in}^2)} = 0.659\ \text{in}$$

Length is way below the width of the hub,





Shaft section at keyseat



Hub section at keyseat

Example:

Specify the complete key geometry for the following system:
½-inch shaft transmits 5 hp at 1750 rpm. The shaft is made from cold-drawn 1040 steel; hub is made from hot-rolled 1213 steel; and a cold-drawn 1020 steel key is to be used. If direction is regularly reversed, determine length of square key required for a SF = 2.5:

Material properties:

- CD 1040 $S_y = 71$ ksi
- HR 1213 $S_y = 58$ ksi
- CD 1020 $S_y = 51$ ksi , $S_s = S_y/2.5$

Solution

$$T = \frac{63{,}000\ hp}{n}$$

$$T = \frac{63{,}000\ (5)}{1750}$$

$$T = 180\ in - lb$$

From the Table: for 1/2-inch shafting, the key is 1/8 x 1/8 inch:

$$L = \frac{4T}{S_s W D}$$

$$L = \frac{4(180in - lb)x2.5}{51{,}000lb/in^2 . 125in . 5in}$$

$$L = 0.565\ in$$

Example:

An electric motor drives a machine through a V-belt pulley. Motor is rated at 6000 in-lb of torque and has a 1½-inch diameter output shaft. System does reverse direction and stops and starts infrequently.

Design key for this application. Use S_y = 82 ksi, use SF of 2

$$L = \frac{2T}{S_s b D}$$

$$L = \frac{2(6000in - lb)x2}{(82{,}000lb/in^2)(.375in)(1.5in)}$$

$$L = 0.52\ inch$$

(When apply safety factor of 2, use 5/8" length)

SPLINES:

Advantages:



- Can carry higher torque for given diameter (vs keys) or
- Lower stress on attachment (gear)
- Better fit, less vibration (spline integral to shaft so no vibrating key)
- May allow axial motion while reacting torque
- Generally hardened to resist wear

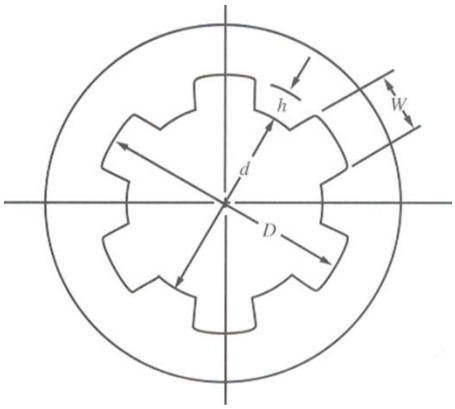
Disadvantage:

- Cost
- Impractical to use as fuse.



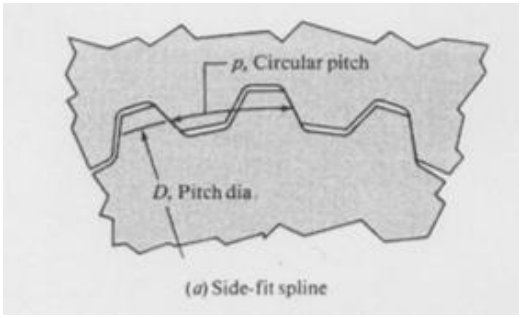


Spline types:



- Straight
  - SAE
  - 4, 6, 10 or 16 splines

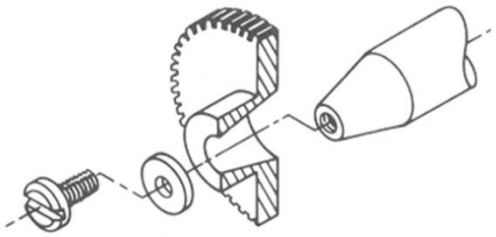
- Involute
  - Pressure angles of 30, 37.5, or 45 deg.
  - Tend to center shafts for better concentricity

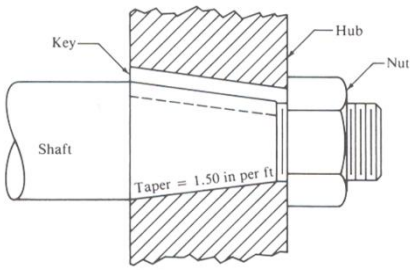


(a) Side-fit spline

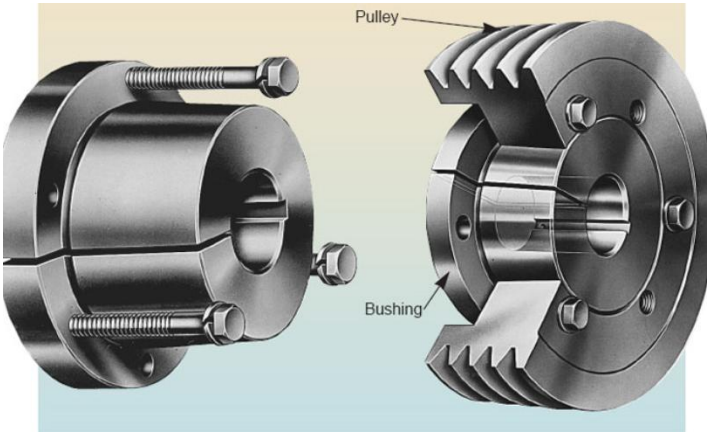
TAPER & SCREW

- Expensive – machining
- Good concentricity
- Moderate torque capacity
- Can use a key too,


(a) Taper and screw



(*b*) Taper and nut



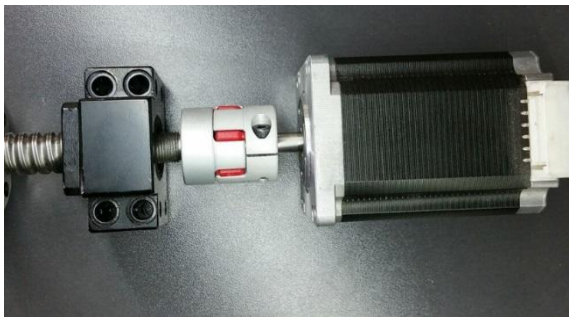


HUMBER

MECH 214 Mechanical Components Design

COUPLING:

- Coupling is used in case where torque needs to be transmitted through shaft where there is potential for misalignment.
- Most cases this happen when separately mounted devices are connected, such as mortar and a pump
- Used to connect two shafts together at their ends to transmit torque from one to the other.
- Two kinds of couplings:
  - RIGID
  - FLEXIBLE

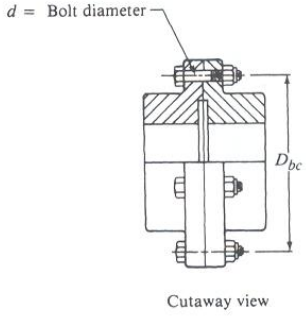


Rigid Coupling:

- NO relative motion between the shafts.
- Precise alignment of the shafts
- Bolts carry torque in shear. N = # of bolts.







Cutaway view

HUMBER

MECH 214 Mechanical Components Design

Flexible Couplings

- Transmit torque smoothly
- Permit some axial, radial and angular misalignment

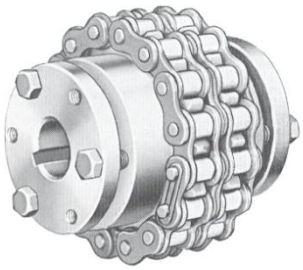
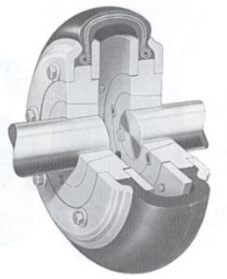


FIGURE 11–16 Chain coupling. Torque is transmitted through a double roller chain. Clearances between the chain and the sprocket teeth on the two coupling halves accommodate misalignment. (Emerson Power Transmission Corporation, Ithaca, NY)

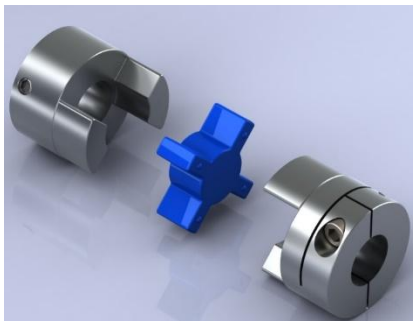


FIGURE 11–17 Ever-Flex coupling. The features of this coupling are that it (1) generally minimizes torsional vibration; (2) cushions shock loads; (3) compensates for parallel misalignment up to 1/32 in; (4) accommodates angular misalignment of ±3°; and (5) provides adequate end float, ±1/32 in. (Emerson Power Transmission Corporation, Ithaca, NY)







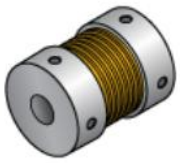












☐ With Slit ☐ Disc Type ☐ Oldham Type ☐ Jaw Type ☐ Bellows Type

UNIVERSAL JOINTS:

- Large shaft misalignments permissible
- Key factors in selection are Torque, Angular Speed and the Operating Angle
- The speed of the output shaft will vary depending upon the misalignment angle and position of the yoke ear.



FIGURE 11–25 Universal joint components
(Curtis Universal Joint Co., Inc., Springfield, MA)

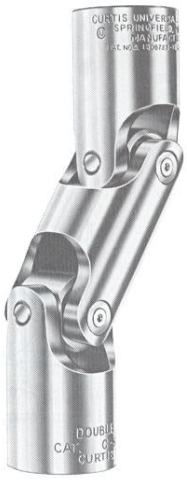
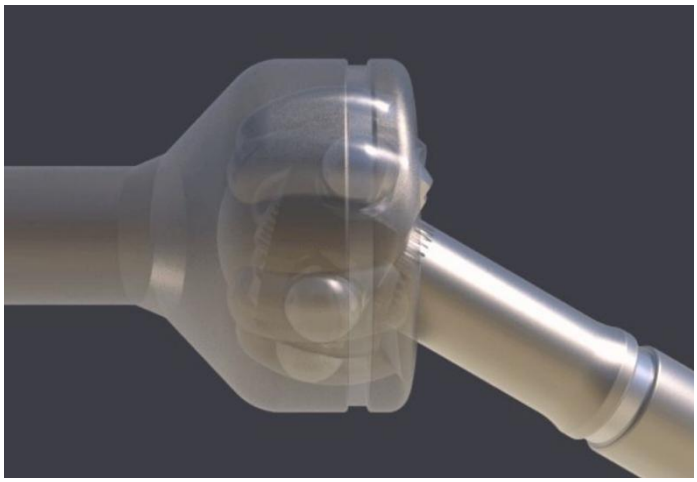


FIGURE 11–26 Double universal joint
(Curtis Universal Joints Co., Inc., Springfield, MA)

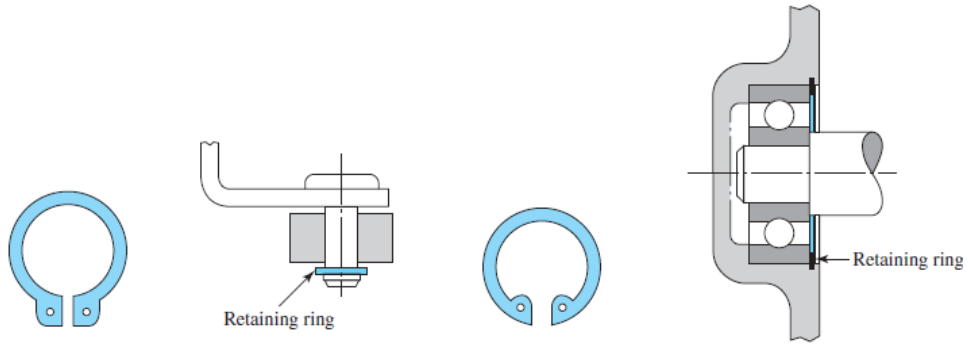
CONSTANT SPEED JOINT:



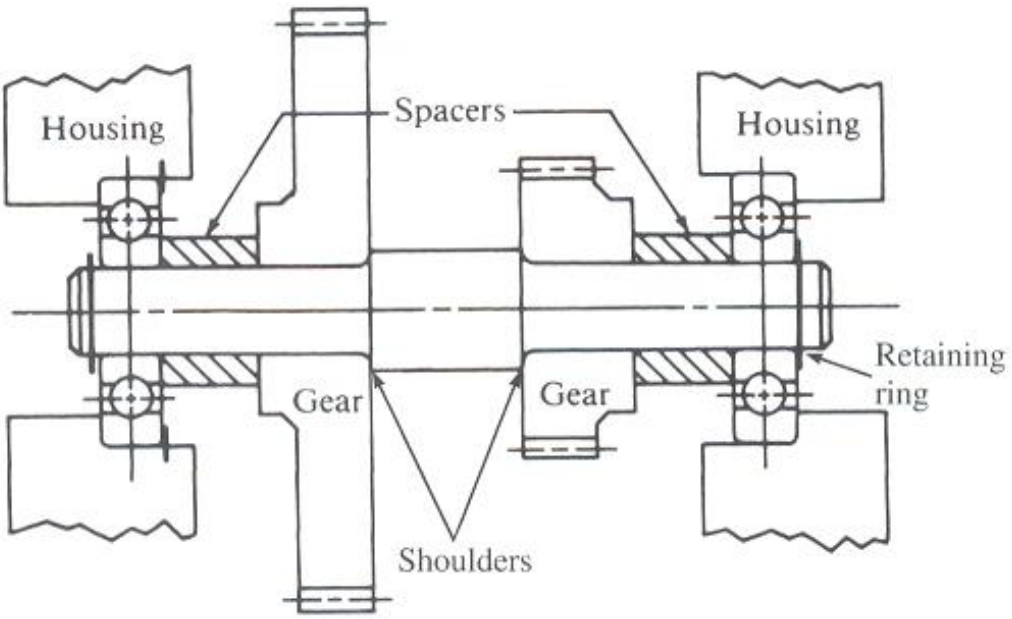


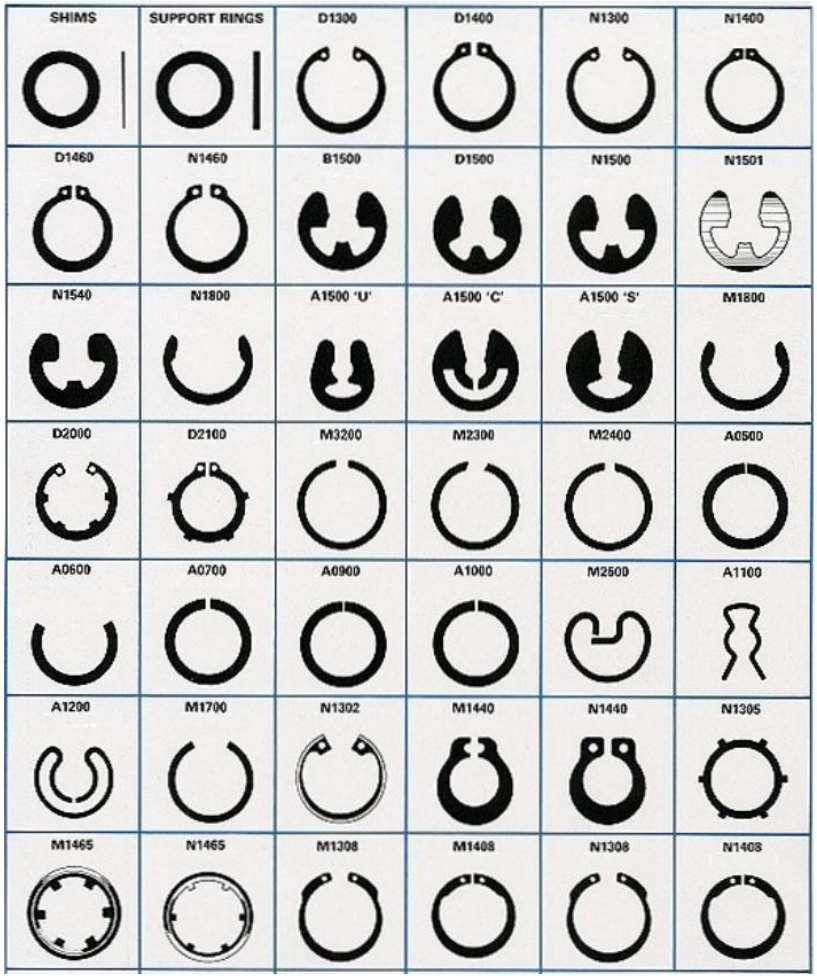
RETAINING RINGS

- A retaining ring is frequently used instead of a shaft shoulder or a sleeve to axially position a component on a shaft or in a housing bore. As shown,



- A groove is cut in the shaft or bore to receive the spring retainer.
- For sizes, dimensions, and axial load ratings, the manufacturers' catalogs should be consulted.
- For the rings to seat nicely at the bottom of the groove, and support axial loads against the sides of the groove, the radius in the bottom of the groove must be reasonably sharp, typically about one-tenth of the groove's width.
- Care should be taken in using retaining rings, particularly in locations with high bending stresses

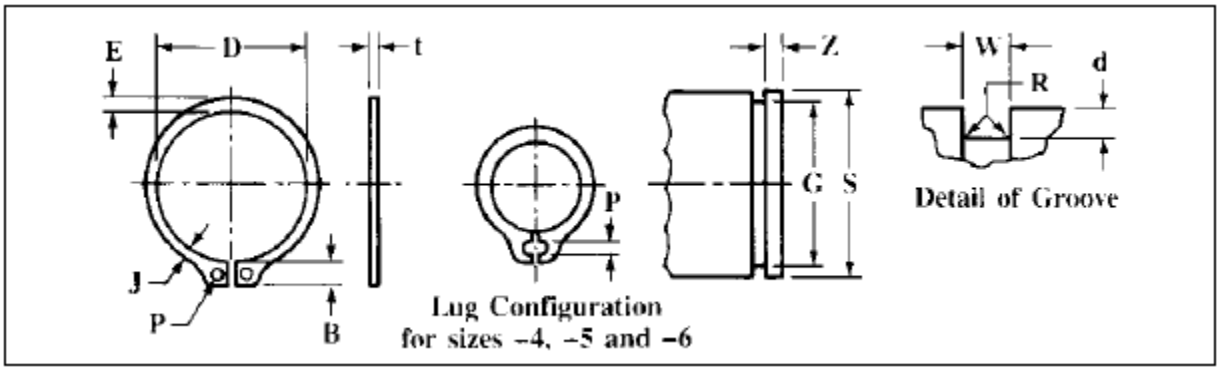



SHIMS
SUPPORT RINGS
D1300
D1400
N1300
N1400
D1460
N1460
B1500
D1500
N1500
N1501
N1540
N1800
A1500 'U'
A1500 'C'
A1500 'S'
M1800
D2000
D2100
M3200
M2300
M2400
A0500
A0600
A0700
A0900
A1000
M2500
A1100
A1200
M1700
N1302
M1440
N1440
N1305
M1465
N1465
M1308
M1408
N1308
N1408

HUMBER

MECH 214 Mechanical Components Design

American National Standard Metric Tapered Retaining Rings —Basic External Series — 3AM1
ANSI B27.7-1977, R1993

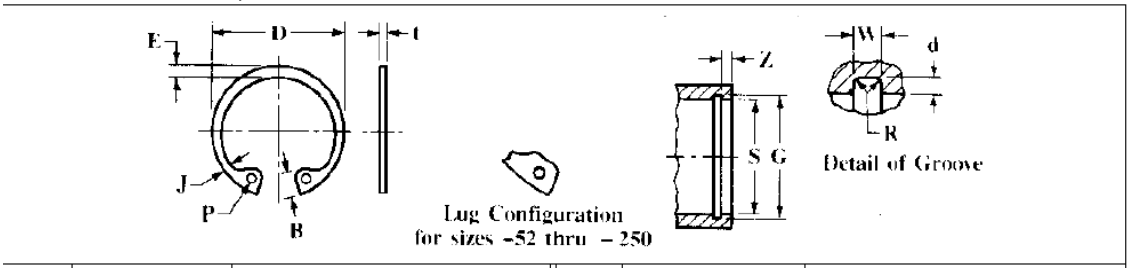


Shaft Dia.	Ring		Groove				Shaft Diam	Ring		Groove			
	Free Dia.	Thick ness	Dia.	Width	Depth	Edge Margin		Free Dia.	Thick ness	Dia.	Width	Depth	Edge Margin
S	*D*	*t*	*G*	*W*	*d* ref	*Z* min	*S*	*D*	*t*	*G*	*W*	*d* ref	*Z* min
4	3.60	0.25	3.80	0.32	0.1	0.3	36	33.25	1.3	33.85	1.4	1.06	3.2
5	4.55	0.4	4.75	0.5	0.13	0.4	38	35.20	1.3	35.8	1.4	1.10	3.3
6	5.45	0.4	5.70	0.5	0.15	0.5	40	36.75	1.6	37.7	1.75	1.15	3.4
7	6.35	0.6	6.60	0.7	0.20	0.6	42	38.80	1.6	39.6	1.75	1.20	3.6
8	7.15	0.6	7.50	0.7	0.25	0.8	43	39.65	1.6	40.5	1.75	1.25	3.8
9	8.15	0.6	8.45	0.7	0.28	0.8	45	41.60	1.6	42.4	1.75	1.30	3.9
10	9.00	0.6	9.40	0.7	0.30	0.9	46	42.55	1.6	43.3	1.75	1.35	4.0
11	10.00	0.6	10.35	0.7	0.33	1.0	48	44.40	1.6	45.2	1.75	1.40	4.2
12	10.85	0.6	11.35	0.7	0.33	1.0	50	46.20	1.6	47.2	1.75	1.40	4.2
13	11.90	0.9	12.30	1.0	0.35	1.0	52	48.40	2.0	49.1	2.15	1.45	4.3
14	12.90	0.9	13.25	1.0	0.38	1.2	54	49.9	2.0	51.0	2.15	1.50	4.5
15	13.80	0.9	14.15	1.0	0.43	1.3	55	50.6	2.0	51.8	2.15	1.60	4.8
16	14.70	0.9	15.10	1.0	0.45	1.4	57	52.9	2.0	53.8	2.15	1.60	4.8
17	15.75	0.9	16.10	1.0	0.45	1.4	58	53.6	2.0	54.7	2.15	1.65	4.9
18	16.65	1.1	17.00	1.2	0.50	1.5	60	55.8	2.0	56.7	2.15	1.65	4.9
19	17.60	1.1	17.95	1.2	0.53	1.6	62	57.3	2.0	58.6	2.15	1.70	5.1
20	18.35	1.1	18.85	1.2	0.58	1.7	65	60.4	2.0	61.6	2.15	1.70	5.1
21	19.40	1.1	19.80	1.2	0.60	1.8	68	63.1	2.0	64.5	2.15	1.75	5.3
22	20.30	1.1	20.70	1.2	0.65	1.9	70	64.6	2.4	66.4	2.55	1.80	5.4
23	21.25	1.1	21.65	1.2	0.67	2.0	72	66.6	2.4	68.3	2.55	1.85	5.5
24	22.20	1.1	22.60	1.2	0.70	2.1	75	69.0	2.4	71.2	2.55	1.90	5.7
25	23.10	1.1	23.50	1.2	0.75	2.3	78	72.0	2.4	74.0	2.55	2.00	6.0
26	24.05	1.1	24.50	1.2	0.75	2.3	80	74.2	2.4	75.9	2.55	2.05	6.1
27	24.95	1.3	25.45	1.4	0.78	2.3	82	76.4	2.4	77.8	2.55	2.10	6.3
28	25.80	1.3	26.40	1.4	0.80	2.4	85	78.6	2.4	80.6	2.55	2.20	6.6
30	27.90	1.3	28.35	1.4	0.83	2.5	88	81.4	2.8	83.5	2.95	2.25	6.7
32	29.60	1.3	30.20	1.4	0.90	2.7	90	83.2	2.8	85.4	2.95	2.30	6.9
34	31.40	1.3	32.00	1.4	1.00	3.0	95	88.1	2.8	90.2	2.95	2.40	7.2
35	32.30	1.3	32.90	1.4	1.05	3.1	100	92.5	2.8	95.0	2.95	2.50	7.5

HUMBER

MECH 214 Mechanical Components Design

American National Standard Metric Tapered Retaining Rings — Basic Internal Series — 3BM1
ANSI B27.7-1977, R1993



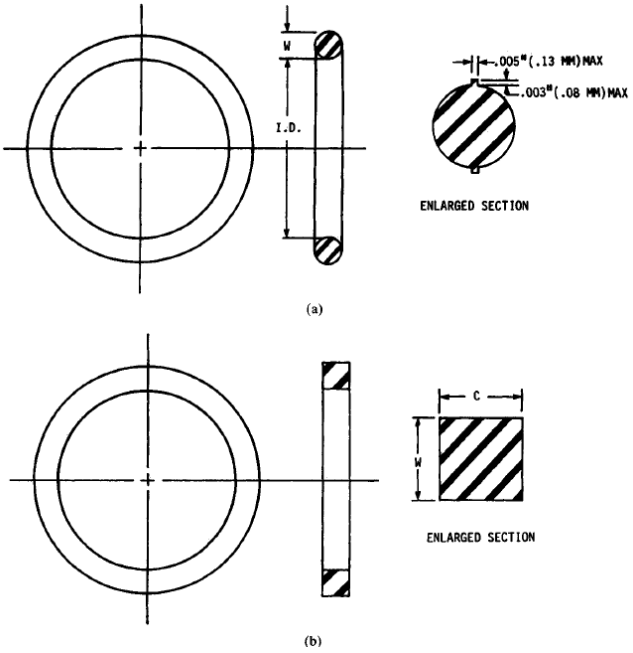
Shaft Dia.	Ring		Groove				Shaft Dia.	Ring		Groove			
	Free Dia.	Thickness	Dia.	Width	Depth	Edge Margin		Free Dia.	Thickness	Dia.	Width	Depth	Edge Margin
S	*D*	*t*	*G*	*W*	*d* ref	*Z* min	*S*	*D*	*t*	*G*	*W*	*d* ref	*Z* min
8	8.80	0.4	8.40	0.5	0.2	0.6	65	72.2	2.4	69.0	2.55	2.00	6.0
9	10.00	0.6	9.45	0.7	0.23	0.7	68	75.7	2.4	72.2	2.55	2.10	6.3
10	11.10	0.6	10.50	0.7	0.25	0.8	70	77.5	2.4	74.4	2.55	2.20	6.6
11	12.20	0.6	11.60	0.7	0.3	0.9	72	79.6	2.4	76.5	2.55	2.25	6.7
12	13.30	0.6	12.65	0.7	0.33	1.0	75	83.3	2.4	79.7	2.55	2.35	7.1
13	14.25	0.9	13.70	1.0	0.35	1.1	78	86.8	2.8	82.8	2.95	2.40	7.2
14	15.45	0.9	14.80	1.0	0.40	1.2	80	89.1	2.8	85.0	2.95	2.50	7.5
15	16.60	0.9	15.85	1.0	0.43	1.3	82	91.1	2.8	87.2	2.95	2.60	7.8
16	17.70	0.9	16.90	1.0	0.45	1.4	85	94.4	2.8	90.4	2.95	2.70	8.1
17	18.90	0.9	18.00	1.0	0.50	1.5	88	97.9	2.8	93.6	2.95	2.80	8.4
18	20.05	0.9	19.05	1.0	0.53	1.6	90	100.0	2.80	95.7	2.95	2.85	8.6
19	21.10	0.9	20.10	1.0	0.55	1.7	92	102.2	2.8	97.8	2.95	2.90	8.7
20	22.25	0.9	21.15	1.0	0.57	1.7	95	105.6	2.8	101.0	2.95	3.00	9.0
21	23.30	0.9	22.20	1.0	0.60	1.8	98	109.0	2.8	104.2	2.95	3.10	9.3
22	24.40	1.1	23.30	1.2	0.65	1.9	100	110.7	2.8	106.3	2.95	3.15	9.5
23	25.45	1.1	24.35	1.2	0.67	2.0	102	112.4	2.8	108.4	2.95	3.20	9.6
24	26.55	1.1	25.4	1.2	0.70	2.1	105	115.8	2.8	111.5	2.95	3.25	9.8
25	27.75	1.1	26.6	1.2	0.80	2.4	108	119.2	2.8	114.6	2.95	3.30	9.9
26	28.85	1.1	27.7	1.2	0.85	2.6	110	120.8	2.8	116.7	2.95	3.35	10.1
27	29.95	1.3	28.8	1.4	0.90	2.7	115	126.0	2.8	121.9	2.95	3.45	10.4
28	31.10	1.3	29.8	1.4	0.90	2.7	120	132.4	2.8	127.0	2.95	3.50	10.5
30	33.40	1.3	31.9	1.4	0.95	2.9	125	137.1	2.8	132.1	2.95	3.55	10.7
32	35.35	1.3	33.9	1.4	0.95	2.9	130	142.5	2.8	137.2	2.95	3.60	10.8
34	37.75	1.3	36.1	1.4	1.05	3.2	135	148.5	3.2	142.3	3.40	3.65	11.0
35	38.75	1.3	37.2	1.4	1.10	3.3	140	154.1	3.2	147.4	3.40	3.70	11.1
36	40.00	1.3	38.3	1.4	1.15	3.5	145	159.5	3.2	152.5	3.40	3.75	11.3
37	41.05	1.3	39.3	1.4	1.15	3.5	150	164.5	3.2	157.6	3.40	3.80	11.4
38	42.15	1.3	40.4	1.4	1.20	3.6	155	168.8	3.2	162.7	3.40	3.85	11.6
40	44.25	1.6	42.4	1.75	1.20	3.6	160	175.1	4.0	167.8	4.25	3.90	11.7
42	46.60	1.6	44.5	1.75	1.25	3.7	165	180.3	4.0	172.9	4.25	3.95	11.9
45	49.95	1.6	47.6	1.75	1.30	3.9	170	185.6	4.0	178.0	4.25	4.00	12.0
46	51.05	1.6	48.7	1.75	1.35	4.0	175	191.3	4.0	183.2	4.25	4.10	12.3
47	52.15	1.6	49.8	1.75	1.40	4.2	180	196.6	4.0	188.4	4.25	4.20	12.6
48	53.30	1.6	50.9	1.75	1.45	4.3	185	202.7	4.8	193.6	5.10	4.30	12.9
50	55.35	1.6	53.1	1.75	1.55	4.6	190	207.7	4.8	198.8	5.10	4.40	13.2
52	57.90	2.0	55.3	2.15	1.65	5.0	200	217.8	4.8	209.0	5.10	4.50	13.5
55	61.10	2.0	58.4	2.15	1.70	5.1	210	230.3	4.8	219.4	5.10	4.70	14.1
57	63.25	2.0	60.5	2.15	1.75	5.3	220	240.5	4.8	230.0	5.10	5.00	15.0
58	64.4	2.0	61.6	2.15	1.80	5.4	230	251.4	4.8	240.6	5.10	5.30	15.9
60	66.8	2.0	63.8	2.15	1.90	5.7	240	262.3	4.8	251.0	5.10	5.50	16.5
62	68.6	2.0	65.8	2.15	1.90	5.7	250	273.3	4.8	261.4	5.10	5.70	17.1
63	69.9	2.0	66.9	2.15	1.95	5.9	…	…	…	…	…	…	…

ELASTOMERIC SEAL RINGS: (O-Ring):

- Seal rings of the O-ring type are used as both static and dynamic seals.
- Static seals: serve the same purpose as gaskets; that is, they provide a seal between two members that are not intended to undergo relative motion.
- Dynamic seals: are used where rotating or reciprocating motion is intended to occur.
- The size is designated by the cross-sectional diameter and the nominal inside diameter (ID).

Standard Sizes of O-Rings

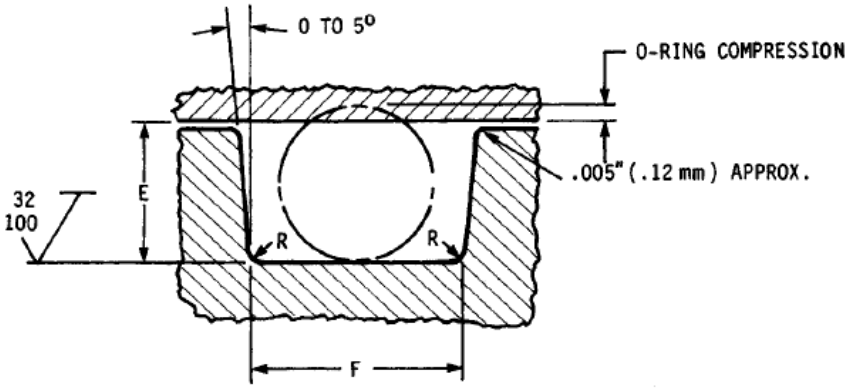
Size no.	w, in	Actual ID, in	
		Minimum	Maximum
006 to 045	0.070 ± 0.003	0.114 ± 0.005	3.989 ± 0.015
110 to 163	0.103 ± 0.003	0.362 ± 0.005	5.987 ± 0.023
210 to 281	0.139 ± 0.004	0.734 ± 0.006	14.984 ± 0.060
325 to 349	0.210 ± 0.005	1.475 ± 0.010	4.475 ± 0.010
425 to 460	0.275 ± 0.006	4.475 ± 0.015	15.475 ± 0.015



Standard Sizes of Rectangular-Section Rings:

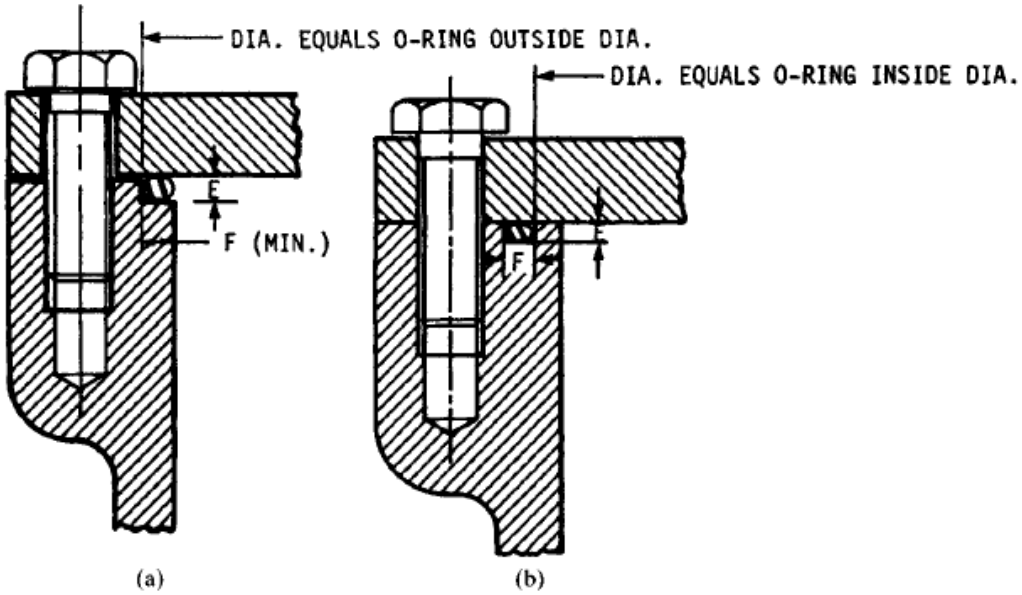
Size no.	w, in	c, in	Actual ID, in	
			Minimum	Maximum
R006 to R045	0.066 ± 0.004	0.066 ± 0.003	0.114 ± 0.005	3.989 ± 0.015
R110 to R163	0.099 ± 0.004	0.099 ± 0.003	0.362 ± 0.005	5.987 ± 0.023
R210 to R281	0.134 ± 0.004	0.134 ± 0.004	0.734 ± 0.006	14.984 ± 0.060
R325 to R349	0.203 ± 0.005	0.203 ± 0.005	1.475 ± 0.010	4.475 ± 0.015
R425 to R460	0.265 ± 0.005	0.265 ± 0.005	4.475 ± 0.015	15.475 ± 0.030

Shape of groove for seal rings.



- A static seal ring application in which the joint is subject to internal pressure only is shown below at (a):

- The groove design in the figure (b) is for a joint subject to external pressure or internal vacuum only.



Seal for rotary motion (dynamic):

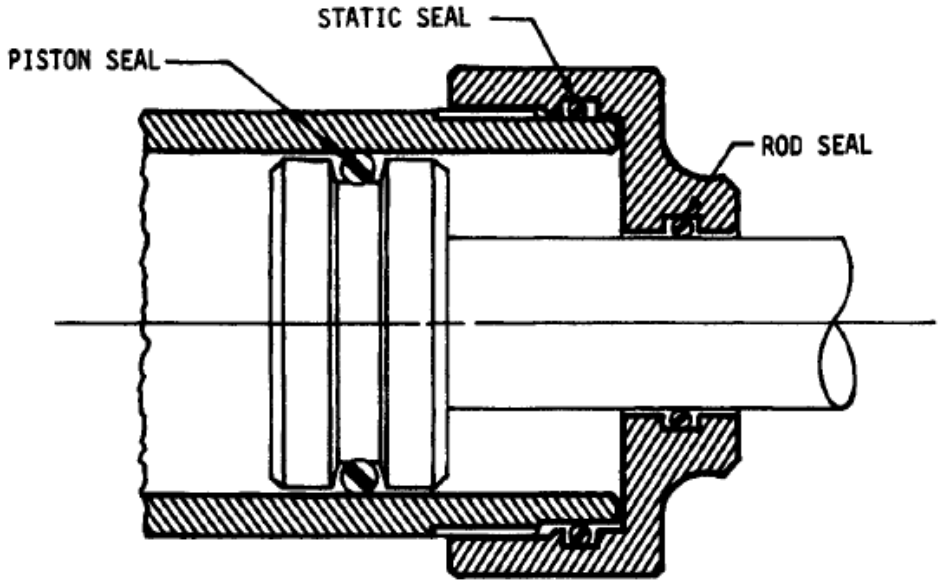
- Seals are required on rotating shafts to retain working fluids, to retain lubricants, and to exclude dirt.
- The selection of a seal type depends on fluid pressure, shaft speed, and whether any leakage can be permitted.

General facts on O-Ring:

- O-rings have been used successfully on rotating shafts when they are installed under compression by using a smaller-than-normal groove diameter in the housing.
- Recommended O-ring cross sections are 0.139 in (3.53 mm) for speeds up to 400 ft/min (2.0 m/s),
- 0.103 in (2.62 mm) for speeds from 400 to 600 ft/min (2.0 to 3.0 m/s),
- and 0.070 in (1.78 mm) for speeds exceeding 600 ft/min (3.0 m/s)

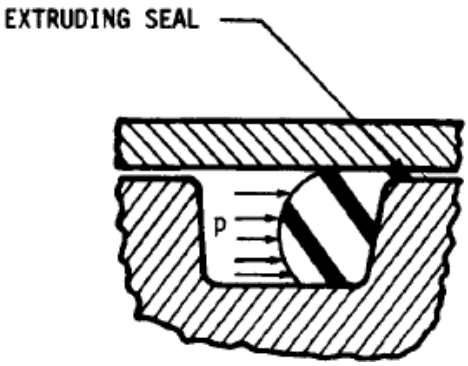
Seals for reciprocating motion and high pressure:

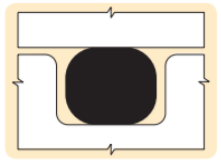
- The O-ring is used extensively because of the low installed cost and effectiveness as a seal.
- It is well adapted to sealing reciprocating motion as well as for use as a static seal.

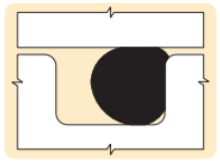


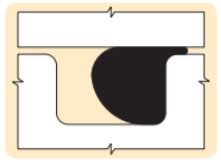
- The amount of clearance that can be permitted depends on the pressure differential across the O-ring and the ring hardness.
- If the clearance is too great, the O-ring is extruded into the clearance space,

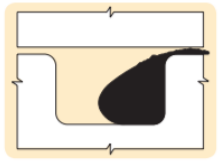
- The reciprocating motion then tears away small pieces of the O-ring, which results in a leaking seal and contamination of the working fluid by the O-ring particles,



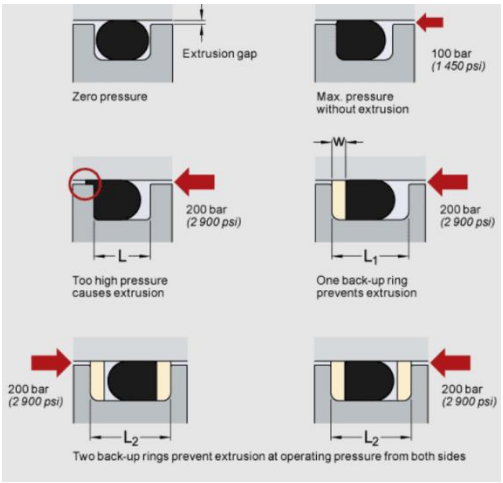

Figure 1-4: O-Ring Installed

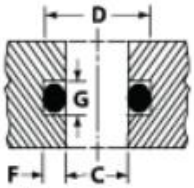

Figure 1-5: O-Ring Under Pressure


Figure 1-6: O-Ring Extruding

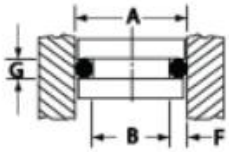

Figure 1-7: O-Ring Failure

- The pressure limitations of O-rings can be overcome by the use of BACKUP RING to prevent O-ring extrusion.
- Backup rings are made from leather, plastics, or metal.
- The metal rings are split like a piston ring for radial compression during assembly into the cylinder.
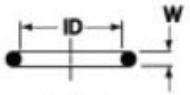




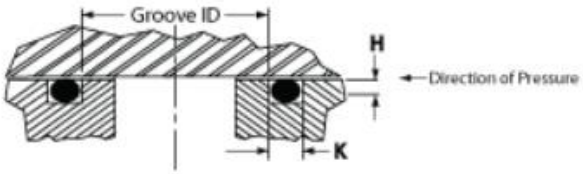
Rod Seal



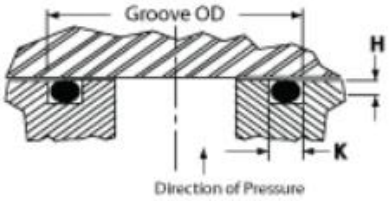
Piston Seal



O-Ring



Face Seal, External Pressure

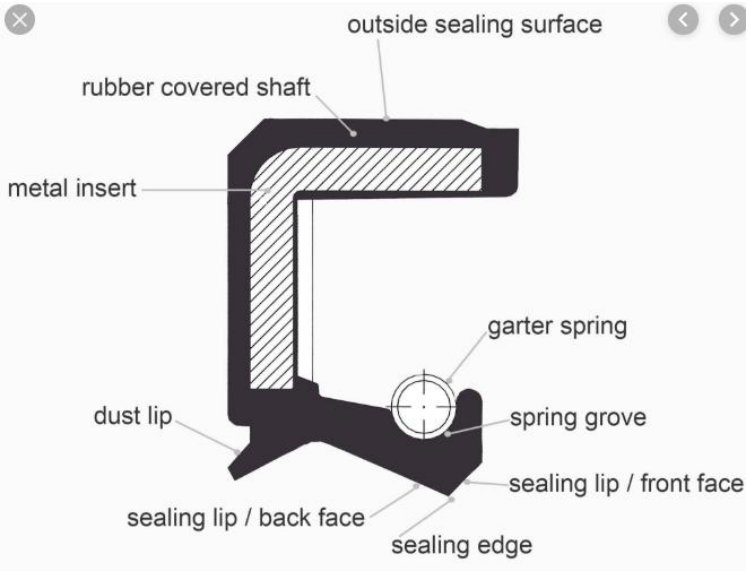


Face Seal, Internal Pressure

Uniform Dash Number	Actual Cross Section Diameter	Diametrical Squeeze (Minimum)		Gland Depth F		Groove Width G +/- .003			Flange Groove		Diametrical Clearance (Maximum) D		Eccentricity[3]	Radius (R)
		Dynamic	Static	Dynamic +.000 -.001	Static +.000 -.004	No Backup Ring	One Backup Ring	Two Backup Rings	Depth H	K min	500 PSI	1500 PSI		
- 001	.040 + .003	.004	.006	.033	.031	.056	-	-	.028/.032	.068	.005	.0025	.002	.010
- 002	.050 + .003	.005	.008	.042	.039	.070	-	-	.037/.041	.078	.006	.003	.002	.010
- 003	.060 + .003	.006	.009	.051	.048	.084	-	-	.045/.050	.091	.007	.0035	.002	.016
- 004 thru –050	.070 + .003	.007	.011	.056	.051	.098	.140	.207	.051/.061	.095	.008	.004	.002	.016
102 thru –178	.103 + .003	.010	.015	.090	.082	.144	.173	.240	.081/.091	.140	.009	.004	.002	.016
-201 thru – 284	.139 + .004	.014	.021	.121	.114	.195	.210	.277	.110/.120	.190	.010	.006	.003	.031
-309 thru –395	.210 + .005	.021	.032	.184	.173	.294	.313	.412	.170/.180	.280	.011	.007	.004	.031
-425 thru – 475	.275 + .006	.028	.042	.241	.227	.385	.410	.540	.231/.241	.370	.012	.008	.005	.047

RADIAL LIP SEALS:

- A section through a radial lip seal is shown below, this type is used primarily for retention of lubricants and exclusion of dirt.
- It is suited for conditions of low lubricant pressure, moderate shaft speeds, less-than-severe environmental conditions and situations where slight leakage may be permitted.
- Radial lip seals are compact, effective, inexpensive, and easily installed.
- The outer case is held in the bearing housing by an interference fit.



- The garter spring provides a uniform radial force to maintain contact between the elastomeric sealing lips and the shaft.
- Lubricant leakage is reduced when hydrodynamic sealing lips are used. Such lips have very shallow grooves molded into the primary sealing lip to pump lubricant out of the contact area. Hydrodynamic sealing lips are manufactured for rotation in one direction only or for rotation in either direction

